

OPEN-SOURCE FULL-STACK WEB DEVELOPMENT WITH AI INTEGRATION

Develops Open-Source full-stack web developers across PHP/Laravel and JavaScript (Node.js, NestJS, React, Vue, GraphQL), with MySQL/MongoDB, REST APIs, Linux, Docker, and Kubernetes, plus open-source ERP and CMS customization. A code-first generative-AI strand (LLM integration, RAG, agents, LangChain/LangGraph) lets graduates add intelligent features. Graduates suit PHP/Laravel Developer, Full-Stack Developer, and Web/Backend Developer roles (with open-source ERP/CMS as adjacent options), and AI application integration as a secondary capability, once they consolidate on a primary stack. Not a machine-learning qualification.

JOB PROFILES

PHP / Laravel Developer · Full-Stack Developer (PHP + JS, Python + JS or MERN) ·
Web Developer / Backend Developer (Node.js/PHP/Django/FastAPI) ·
Open-Source ERP (Odoo) Developer · CMS Developer ·
AI Application Developer (LangChain/LangGraph) - (Generative AI — LLM · RAG · Agentic) ·
Workflow Automation Specialist



TOTAL HOURS

1,350hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



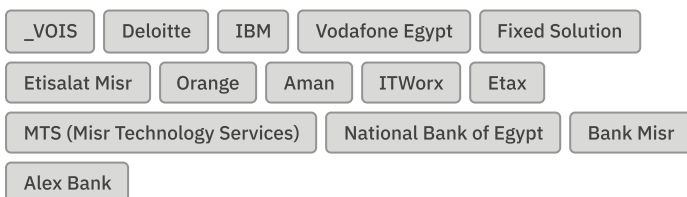
CERTIFICATIONS

3

Industry-certified



Where graduates work



Certifications earned

- RHCSA** — Red Hat Certified System Administrator
- CLF-C02** — AWS Certified Cloud Practitioner
- MongoDB Associate Developer



Selection process

01 IQ & English Exam

- Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam

- Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview

- Interview with the technical committee.

04 Interpersonal Interview

- Candidates who pass the technical interview are invited to this final stage.



Who may apply

All University Graduates, No special requirements needed to apply to the program. The applicants must have a first degree from a recognized university or institution of higher education.

Program content – 1,350 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Software Engineering Foundations	291 hrs	22%	Introduction to Software Engineering & UML, Agile Software Development Methodologies, Object-Oriented Programming, Data Structures and Algorithms, Database Fundamentals, Computer Network Fundamentals, Operating Systems Fundamentals, Client-Side Technologies
● Application Development	438 hrs	32%	ReactJS, Angular, Node.js, Next.js, VueJS, NestJS, Python Programming, Django, FastAPI, PHP Programming, Core Java Programming, Web APIs, Push / Pull Technologies, MySQL Database, MongoDB, Open-Source ERP Systems, Develop AI Plugins, Laravel
● AI-Powered Software Engineering	102 hrs	8%	Introduction to Generative AI and Prompt Engineering, Gen-AI Development Foundation, AI-Assisted Coding and AI Coding Agents, AI Workflow Automation and App Orchestration, Advanced Gen-AI Programming: RAG and Agentic Software, LangChain & LangGraph
● Cloud, DevOps & Secure Software Delivery	162 hrs	12%	Cloud Computing and Docker Fundamentals, Kubernetes, CI/CD Basics, Version Control Systems, Bash Shell Script, Red Hat System Administration I, Apache & Nginx Web Servers, Design Patterns, Unit & Automated Testing, Application Security Basics
● Soft Skills	57 hrs	4%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
● Graduation Project	300 hrs	22%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Each Student must deliver at least ONE Freelancing job and Complete the Graduation Project.

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Programming Fundamentals
<https://www.edx.org/learn/computer-science/harvard-university-cs50-s-introduction-to-computer-science>
- OOP Fundamentals
<https://maharatech.gov.eg/course/view.php?id=2206>
- Web Fundamentals
<https://maharatech.gov.eg/course/view.php?id=36>
- Database Fundamentals
<https://maharatech.gov.eg/course/view.php?id=740>
- Linux Fundamental
<https://maharatech.gov.eg/course/view.php?id=2155>
- ITI Values & Mission
<https://iti.gov.eg/about-us>

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.